

PEC-Nav: Self-Calibrating Frontier Search for Embodied Question Answering

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Abstract

Frontier-based navigation agents evaluate every frontier with the same static scoring function throughout an episode, regardless of whether their decisions prove correct. We propose PEC-Nav, a self-calibrating search mechanism for Embodied Question Answering that learns from its own prediction errors online. Before exploring a frontier, the agent predicts the semantic layout beyond it; after arriving, it measures the divergence between prediction and observation as a surprise signal. High surprise broadens exploration; low surprise narrows the search toward the goal—an adaptive shift that emerges automatically without environment-specific training. PEC-Nav wraps this Predict-Explore-Correct loop around a language-driven EQA pipeline with episodic memory and follow-up QA. Because memory and reasoning modules are held fixed, any improvement in answer quality is directly attributable to search. On LMEE-Bench (GOAT-Bench Val Unseen, 345 episodes), PEC-Nav achieves 29.74% SR and 20.88 SPL using a 4B-parameter VLM on a single consumer GPU. Disabling the correction loop degrades performance to 12.75% SR; re-enabling it yields a 133% relative improvement—confirming that prediction errors, not predictions, drive effective search.

1. Introduction

Every frontier-based navigation agent faces the same dilemma: it must decide where to explore next using a scoring function that knows nothing about whether its previous decisions were any good. A frontier evaluated at step one is assessed with the same model, the same confidence, and the same exploration-exploitation balance as one evaluated at step two hundred—regardless of how many wrong rooms the agent has already walked into. The search strategy is *static*, and in Embodied Question Answering (EQA), where the agent must navigate an unfamiliar 3D environment to gather evidence for answering a natural language question, this rigidity is especially costly. The agent must jointly solve *what* to search for, *where* to look, and *how long* to keep

exploring, yet it has no mechanism to learn from its own mistakes within an episode.

Recent memory-augmented methods [8, 18, 21] have made impressive progress on the reasoning side of EQA, showing that richer memory representations lead to better answers. But these methods inherit whatever observations navigation provides. If the agent wastes steps exploring irrelevant rooms, the memory is impoverished no matter how sophisticated the retrieval. *The quality of reasoning is bounded by the quality of search.*

We introduce a simple mechanism that breaks this limitation: *predict before you explore, then learn from your mistakes.* Before moving through a frontier, the agent uses a vision-language model to predict the semantic layout beyond it—what room type and object categories are likely to appear. After arriving, it compares this prediction against what it actually observes. The discrepancy, measured as KL divergence, produces a *surprise signal* that reveals how well the agent understands the current environment. High surprise means the agent’s model is miscalibrated; the rational response is to broaden exploration. Low surprise means the model has converged; the rational response is to narrow the search toward the goal. This exploration-exploitation shift emerges automatically from the surprise dynamics—no explicit scheduling, no environment-specific training.

This predict-then-correct loop is made practical by the capabilities of modern vision-language models, which can produce meaningful semantic layout predictions from a single egocentric frontier view. Even when these predictions are imperfect—and they frequently are—the *pattern of errors* is informative: systematic misprediction in one environment tells the agent something that no individual prediction can.

We instantiate this idea in **PEC-Nav** (Predict-Explore-Correct Navigation), a language-driven EQA pipeline whose search module forms a closed feedback loop. Given a natural language instruction, the pipeline extracts a structured navigation goal via LLM parsing, then applies the PEC loop: a semantic predictor anticipates the layout beyond each frontier, an uncertainty-aware policy selects frontiers by balancing goal relevance against prediction confidence, and a

correction module updates both the predictor’s confidence scaling and the policy’s exploration-exploitation weights online (Fig. 1). Observations gathered during navigation populate an episodic memory from which the agent answers follow-up questions. Because the memory and QA modules are held constant across all experiments, any change in answer quality is directly attributable to search quality—making this pipeline a controlled testbed for the hypothesis that *search is the bottleneck in embodied reasoning*.

We evaluate PEC-Nav on LMEE-Bench (GOAT-Bench Val Unseen, 36 scenes, 345 episodes). Using a 4B-parameter VLM on a single consumer GPU, PEC-Nav achieves 29.74% SR and 20.88 SPL, competitive with methods that rely on models orders of magnitude larger. The correction ablation provides the sharpest evidence for our thesis: disabling the feedback loop while keeping the predictor active *degrades* performance to 12.75% SR—worse than not predicting at all. Re-enabling correction yields a 133% relative improvement, confirming that it is prediction *errors*, not predictions, that drive effective search.

Our contributions are:

- **The Predict-Explore-Correct loop**, the first self-calibrating frontier search mechanism for embodied navigation. The agent predicts semantic layouts beyond frontiers, measures surprise upon arrival, and uses accumulated surprise to adapt both predictor confidence and exploration-exploitation balance within a single episode.
- **A diagnostic EQA pipeline** in which memory and reasoning are held fixed, isolating search as the sole experimental variable. This design directly tests—and confirms—the hypothesis that search quality is the bottleneck for downstream embodied reasoning.
- **Empirical validation on LMEE-Bench** showing that self-calibrating search outperforms static alternatives in navigation efficiency and downstream QA, with within-episode calibration dynamics measured directly from correction logs.

2. Related Work

2.1. Frontier-Based Navigation

Frontier-based exploration [20] directs agents toward boundaries between explored and unexplored space. SemExp [4] learns a goal-conditioned policy over semantic maps. CoW [7] and VLFM [24] replace learned scorers with CLIP [13] and VLM-based zero-shot frontier evaluation, respectively. ESC [28] adds LLM-derived commonsense constraints to frontier selection. SG-Nav [23] reasons over online 3D scene graphs, and UniGoal [22] unifies category, image, and text goals into a single evaluation mechanism. SCOPE [17] introduces potential-based scoring with a spatio-temporal graph and a self-reconsideration mechanism for revisiting past decisions.

All of these methods evaluate frontiers with a fixed function throughout the episode. Ours is the first to close the loop: predictions about unseen space are verified upon arrival, and the resulting error signal recalibrates both the predictor and the selection policy online.

2.2. Embodied Question Answering and Memory

EQA [6] requires agents to navigate and gather evidence before answering. Explore-EQA [14] demonstrated that VLM-guided frontier selection outperforms random exploration. GraphEQA [15] and Fine-EQA [9] improved reasoning through scene graphs and hybrid exploration, respectively. ToolEQA [19] augments agents with external tools for multi-step reasoning, while Memory-Centric EQA [25] feeds memory signals back into the planner. CoV [2] introduces chain-of-view prompting for viewpoint selection.

On the memory side, 3D-Mem [8] builds volumetric spatial memories, RA-Mem [21] enables retrieval-augmented recall over long trajectories, and MemoryExplorer [18] structures episodic memory with spatial-temporal annotations for multi-goal exploration on LMEE-Bench [10]—the benchmark we adopt.

These methods improve what the agent does with gathered information. We improve what information gets gathered. The two directions are complementary: our search module can replace the navigation component in any of these pipelines.

2.3. World Models for Navigation

Predicting unseen space before acting is a natural idea. Imagine Before Go [26] trains a generative map to hallucinate unvisited regions. ImagineNav [27] and WMNav [11] prompt VLMs to imagine future states for frontier evaluation. UniWM [5] unifies foresight and planning in a single multi-modal backbone. FrontierNet [16] learns to predict frontier utility from raw observations.

These methods use predictions *for planning*—they imagine desirable states and navigate toward them. We use predictions *for calibration*—the value lies not in what the agent predicts, but in how wrong it turns out to be. This distinction is fundamental: planning from predictions fails silently when predictions are wrong, whereas calibration from prediction errors explicitly detects and adapts to failure.

2.4. Positioning

Table 1 summarizes the landscape. PEC-Nav is the only method that combines adaptive search, language-driven goals, follow-up QA, zero-shot operation, and world-model prediction within a single pipeline.

3. Method

We present PEC-Nav, a language-driven pipeline for embodied exploration and question answering. Given a natu-

Table 1. Qualitative comparison across key design dimensions. **Adapt.**: search adapts during episode. **QA**: follow-up question answering. **ZS**: zero-shot. **WM**: world model prediction. PEC-Nav uniquely combines all five capabilities.

Method	Venue	Goal	Search	Adapt.	Memory	QA	ZS	WM	Benchmark
SemExp [4]	NeurIPS’20	Category	Learned policy	×	Semantic map	×	×	×	MP3D
CoW [7]	CoRL’23	Text	CLIP frontier	×	—	×	✓	×	PASTURE
ESC [28]	ICML’23	Text	LLM + frontier	×	—	×	✓	×	HM3D
VLFM [24]	ICRA’24	Text	VLM frontier	×	—	×	✓	×	HM3D
Img.B.Go [26]	CVPR’24	Category	Gen. map	×	Pred. map	×	×	✓	MP3D
Explore-EQA [14]	CVPR’24	Question	VLM frontier	×	—	✓	✓	×	HM-EQA
SG-Nav [23]	NeurIPS’24	Text	3D scene graph	×	Scene graph	×	✓	×	HM3D
GraphEQA [15]	ICRA’25	Question	Scene graph	×	3D graph	✓	✓	×	OpenEQA
Fine-EQA [9]	ICCV’25	Question	Hybrid	×	—	✓	×	×	EXPRESS
UniGoal [22]	CVPR’25	Any	Universal	×	—	×	✓	×	HM3D
ToolEQA [19]	2025	Question	Tool-guided	×	—	✓	×	×	HM-EQA
Mem-Centric [25]	2025	Question	Planner	×	Feedback mem.	✓	×	×	HM-EQA
UniWM [5]	2026	Visual	World model	×	Latent mem.	×	×	✓	GoStanford
SCOPE [17]	AAAI’26	Image	Potential graph	×	Pot. graph	×	✓	×	MP3D
MemExplorer [18]	CVPR’26	Multi	Frontier+LLM	×	Episodic bank	✓	✓	×	LMEE
PEC-Nav (Ours)	—	Question	PEC frontier	✓	Episodic	✓	✓	✓	LMEE

ral language instruction, the pipeline extracts a structured navigation goal, explores using a self-calibrating frontier search, stores observations in episodic memory, and answers follow-up questions from the accumulated memory. Figure 1 provides an overview.

3.1. Problem Setup and Goal Extraction

We address EQA on the GOAT-Bench Val Unseen split [10], comprising 36 indoor scenes from HM3D. Unlike standard ObjectNav, where the goal is a category label or image, our agent receives a natural language instruction (e.g., “Can you find the refrigerator next to the kitchen cabinet?”), which conveys both spatial context and relational cues. We extract a structured goal using GPT-4o [12], which receives *only* the instruction text—no scene geometry or ground-truth labels—and outputs a **target category** (e.g., “refrigerator”) and a **contextual description** (e.g., “likely in a kitchen, near cabinets”). This structured goal is shared by both the VLM frontier scorer and the semantic predictor.

3.2. Base System: SCOPE

Our pipeline builds on SCOPE [17], a zero-shot frontier-based exploration framework. SCOPE constructs a top-down occupancy map from egocentric depth, identifies frontiers as boundaries between explored and unexplored space, and scores each frontier f using a VLM that produces four signals: semantic richness (r_f), explorability (e_f), goal relevance ($g_{r,f}$), and an overall potential estimate (p_f). These scores are organized in a spatio-temporal potential graph with exponential decay ($\gamma = 0.95$) to reflect assessment staleness:

$$s_t = \gamma^{\Delta t} \cdot s_{t-1} \quad (1)$$

We modify SCOPE in exactly one way: goal relevance is computed against our text-based extracted goal rather than a reference image. All other components—frontier detection, VLM scoring, potential graph, self-reconsideration—are retained unchanged. Our contribution begins *after* scoring: we augment frontier *selection* with the PEC loop described next.

3.3. Predict-Explore-Correct Loop

At each decision step, the agent *predicts* what lies beyond each frontier, *explores* the most promising one, and *corrects* its model by comparing prediction against observation. The correction signal feeds back to modulate future predictions and selection.

3.3.1 Predict

For each frontier f , the VLM is prompted with the frontier’s egocentric view and extracted goal to return the top-3 most likely categories $\{c_1, c_2, c_3\}$ with confidences $\hat{p}_i \in [0, 1]$. These are modulated by a self-calibration scale $\sigma_t \in [0.4, 1.0]$:

$$\hat{P}(c_i) = \hat{p}_i \cdot \sigma_t, \quad \hat{P}(\text{other}) = \max\left(\epsilon, 1 - \sum_{i=1}^3 \hat{p}_i \sigma_t\right) \quad (2)$$

where $\epsilon = 10^{-3}$ prevents zero probabilities. When prior predictions have been accurate, $\sigma_t \approx 1.0$ and raw VLM confidences pass through; when predictions have been consistently wrong, σ_t decreases, flattening the distribution toward uniform. The update rule for σ_t is derived in Sec. 3.4. Each $\hat{P}_f$ is stored keyed by frontier identity for later correction.

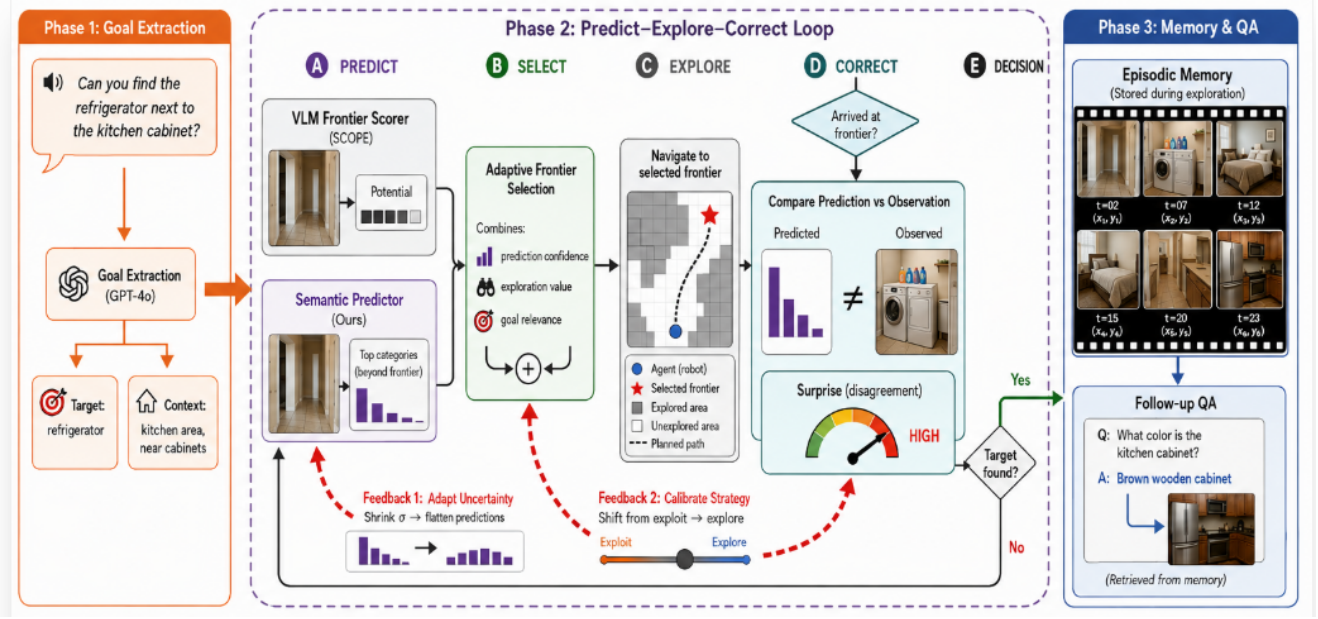


Figure 1. Overview of PEC-Nav. **Phase 1:** A natural language instruction is parsed into a structured goal (target category + spatial context). **Phase 2:** The Predict-Explore-Correct loop drives frontier search—a semantic predictor anticipates layouts beyond frontiers (A), an adaptive policy selects the most informative frontier by combining prediction confidence, exploration value, and goal relevance (B), the agent navigates there (C), and compares prediction against observation to compute a surprise signal (D). Two feedback pathways adapt predictor confidence σ_t (Feedback 1) and exploration-exploitation weights w_e, w_g (Feedback 2) online. **Phase 3:** Observations stored during navigation form an episodic memory for follow-up question answering.

3.3.2 Explore

The agent selects $f^* = \arg \max_f \text{val}'(f)$ (Sec. 3.4) and navigates toward it using SCOPE’s point-goal controller. Ego-centric RGB observations at each step are recorded into an episodic memory buffer with spatial-temporal coordinates.

3.3.3 Correct

When the agent arrives within $R = 3.0$ m of a previously-predicted frontier f , the correction phase fires. The distance is computed on the (x, z) ground plane after transforming the agent’s position from the Habitat frame to the TSDF frame:

$$d = \left\| (w_f^{(x)}, w_f^{(z)}) - (\text{pts}_{\text{norm}}^{(x)}, \text{pts}_{\text{norm}}^{(z)}) \right\|_2 \quad (3)$$

Upon arrival, the agent constructs an observed distribution Q over the same category set $C = \{c_1, c_2, c_3, \text{other}\}$ from semantic detections near f :

$$Q(c) = \frac{\text{count}(c) + \epsilon}{\sum_{c' \in C} (\text{count}(c') + \epsilon)} \quad (4)$$

The surprise signal is the KL divergence between prediction and observation:

$$D_{\text{KL}}(\hat{P} \| Q) = \sum_{c \in C} \hat{P}(c) \log \frac{\hat{P}(c)}{Q(c)} \quad (5)$$

Low D_{KL} indicates the predictor is well-calibrated for this environment; high D_{KL} signals the agent’s model is unreliable and search should adapt.

3.4. Self-Calibration

The surprise signal drives two feedback loops through a shared running estimate.

Running surprise. We maintain an EMA of per-correction KL values:

$$\bar{s}_t = (1 - \beta) \bar{s}_{t-1} + \beta \cdot D_{\text{KL}}^{(t)}, \quad \beta = 0.3, \quad \bar{s}_0 = 0 \quad (6)$$

The initialization $\bar{s}_0 = 0$ encodes an optimistic prior: the agent trusts its predictions until evidence accumulates otherwise. We squash to a bounded signal:

$$n_t = 1 - e^{-\bar{s}_t} \in (0, 1) \quad (7)$$

Adapt: predictor confidence. The calibration signal directly controls σ_t :

$$\sigma_t = 1 - 0.6 n_t \in [0.4, 1.0] \quad (8)$$

This closes the loop: predictions generate surprise, surprise reduces confidence, reduced confidence produces flatter distributions that generate less extreme surprise. The system converges toward a confidence appropriate for the environment.

Calibrate: exploration-exploitation reweighting. The frontier value function shifts along the explore-exploit spectrum:

$$\text{val}(f) = w_p p_f + w_e e_f + w_g g_{r,f} \quad (9)$$

$$w_p = 0.5, \quad w_e = 0.3 + 0.2 n_t, \quad w_g = 0.2 - 0.2 n_t$$

Weights sum to 1.0 for all n_t . When predictions are unreliable ($n_t \rightarrow 1$), w_e reaches 0.5 and w_g drops to zero—the agent broadens exploration. When predictions are reliable ($n_t \approx 0$), the agent exploits toward the goal. A soft visit penalty discourages unproductive revisitation:

$$\text{val}'(f) = \frac{\text{val}(f)}{1 + 0.5 \cdot \text{visits}(f)} \quad (10)$$

Algorithm 1 summarizes the complete PEC-Nav decision cycle.

Algorithm 1 PEC-Nav decision cycle

Require: Frontiers $\mathcal{F}$, goal g , state $(\sigma_t, n_t, \bar{s}_t)$
Ensure: Selected frontier f^* , updated state

- 1: **for** each $f \in \mathcal{F}$ **do**
- 2: $p_f, e_f, g_{r,f} \leftarrow \text{VLMScore}(f, g)$ {SCOPE}
- 3: $\hat{P}_f \leftarrow \text{PredictLayout}(f, g, \sigma_t)$ {Eq. 2}
- 4: $\text{val}'(f) \leftarrow \text{Eq. 9-10}$
- 5: **end for**
- 6: $f^* \leftarrow \arg \max_f \text{val}'(f)$; navigate toward f^*
- 7: **if** $d(f') \leq R$ for any previously-scored f' **then**
- 8: $Q \leftarrow \text{ObservedLayout}(f')$ {Eq. 4}
- 9: $\bar{s}_t \leftarrow (1-\beta)\bar{s}_{t-1} + \beta \cdot D_{\text{KL}}(\hat{P}_{f'} \| Q)$
- 10: $n_t \leftarrow 1 - e^{-\bar{s}_t}$; $\sigma_t \leftarrow 1 - 0.6n_t$
- 11: update w_e, w_g via n_t
- 12: **end if**

3.5. Episodic Memory and QA

During navigation, every egocentric RGB observation is stored with spatial-temporal coordinates. After exploration, follow-up questions are answered by retrieving relevant observations and reasoning over them. The memory and QA modules are identical across all experiments—only search varies—so any change in answer quality is attributable solely to search.

3.6. Implementation Details

All VLM inference uses Qwen3-VL-4B-Instruct [3] in bf16 on a single NVIDIA RTX 3090 (24 GB). Goal extraction uses GPT-4o [12] via API. Ground-truth category distributions for the CORRECT phase are constructed from the Habitat simulator’s semantic sensor, which provides per-pixel labels from HM3D annotations. Table 2 lists all hyperparameters; PEC-Nav-specific values were tuned on 5 held-out scenes and fixed for all experiments.

Table 2. PEC-Nav hyperparameters. †: inherited from SCOPE.

Symbol	Value	Source	Description
γ	0.95	SCOPE†	Temporal score decay
ϵ	10^{-3}	PEC-Nav	Distribution smoothing
R	3.0 m	PEC-Nav	Arrival detection radius
β	0.3	PEC-Nav	Surprise EMA coefficient
$\bar{s}_0$	0	PEC-Nav	Initial running surprise
w_p	0.5	PEC-Nav	Potential weight (fixed)
w_e range	[0.3, 0.5]	PEC-Nav	Explorability weight (adaptive)
w_g range	[0.0, 0.2]	PEC-Nav	Goal-relevance weight (adaptive)
σ range	[0.4, 1.0]	PEC-Nav	Predictor confidence bounds
Visit disc.	0.5	PEC-Nav	Revisitation penalty factor

4. Experiments

We evaluate PEC-Nav through four complementary analyses: a contextual comparison against published methods, a controlled comparison isolating the search module, ablation studies quantifying each component’s contribution, and a self-calibration analysis providing direct evidence of within-episode adaptation.

4.1. Setup

Dataset. All experiments use the GOAT-Bench Val Unseen split [10]: 36 HM3D scenes, 345 episodes with description-type goals (15 of 360 excluded for missing valid descriptions). We evaluate exclusively on description goals—harder than category goals, requiring natural language understanding and spatial reasoning.

Metrics. **Success Rate (SR):** fraction of episodes where the agent finds the target (1.0 m threshold). **SPL** [1]: success weighted by path efficiency, with failures contributing 0. We report both by-snapshot (all 345 episodes) and by-distance (stratified by geodesic distance to target) variants.

Hardware. Qwen3-VL-4B-Instruct in bf16 on a single RTX 3090 (24 GB). Goal extraction via GPT-4o API. No multi-GPU inference, no closed-source VLM for navigation.

4.2. Contextual Comparison

Table 3a situates PEC-Nav against published GOAT-Bench results. An important caveat: most baselines report subtask-level metrics across all goal modalities in multi-goal episodes (5–10 subtasks); we report episode-level metrics on description goals only—a strictly harder setting.

Despite this disadvantage and using a 4B-parameter model, PEC-Nav achieves 29.74% SR and 20.88 SPL. The SR is competitive with SenseAct Skill Chain (29.5%), which uses an RL policy trained on in-domain data. The SPL of 20.88 is the highest among all zero-shot methods—nearly double that of VLMNav (6.6) and DyNaVLM (10.2), both of which use GPT-4V ($\sim 1.8\text{T}$ parameters). This suggests

that *adapting search online* matters more than *scaling the model*.

4.3. Controlled Comparison

The contextual comparison cannot isolate PEC-Nav’s contribution due to differences in evaluation protocol and model backbone. Table 3b presents the definitive test: same pipeline, same Qwen3-VL-4B model, same 345 episodes, same goal extraction, same memory, same QA—only the search module varies. The baseline uses SCOPE’s static frontier scoring; PEC-Nav adds the predict-explore-correct loop.

Any difference in SR or SPL is therefore attributable *solely* to self-calibrating search.

Table 3. Navigation results. (a) Contextual comparison (†: from original publications; PEC-Nav uses description goals only, a harder setting). (b) Controlled comparison: identical pipeline, only search differs.

(a) Contextual comparison					
Method	VLM	Params	ZS	SR (%) ↑	SPL ↑
SenseAct-NN† [10]	RL policy	—	✗	12.3	6.8
CLIP on Wheels† [7]	CLIP	0.4B	✓	16.1	10.4
VLMNav†	GPT-4V	~1.8T	✓	16.3	6.6
Modular GOAT† [10]	CLIP+det.	—	✗	24.9	17.2
DyNavLM†	GPT-4V	~1.8T	✓	25.5	10.2
3D-Mem† [8]	Qwen2.5-VL	7B	✓	28.8	15.8
SenseAct Skill Chain† [10]	RL policy	—	✗	29.5	11.3
PEC-Nav (ours)	Qwen3-VL	4B	✓	29.74	20.88

(b) Controlled comparison			
Search Method	SR (%) ↑	SPL ↑	
SCOPE (static search)	—	—	
PEC-Nav (self-calibrating)	29.74	20.88	

4.4. Ablation Studies

The correction loop is everything. Table 4a presents the sharpest result in this paper. In an early implementation, a coordinate-frame mismatch between the Habitat localizer and the TSDF frontier map caused arrival detection (Eq. 3) to almost never fire. The predictor ran, predictions were stored, but the CORRECT phase never triggered—so surprise was never computed, σ_t never updated, and w_e/w_g never shifted. The agent operated with a static predictor layered on SCOPE: *strictly more computation, no adaptation*.

Result: 12.75% SR—*worse* than SCOPE alone. The predictor without correction is not merely unhelpful; it actively hurts. Fixing the coordinate frame so correction fires reliably produces a **133% relative improvement** (12.75% → 29.74%). This is the cleanest possible evidence that prediction errors, not predictions, are the operative signal.

Component ablation. Table 4b cumulatively adds each component to the SCOPE baseline. Each addition is ex-

pected to contribute: goal extraction provides richer targets, the predictor adds layout-level signal, uncertainty-aware selection introduces adaptivity, and the corrector closes the loop.

Self-calibration dynamics. Table 4c directly measures whether the agent calibrates within an episode. We extract σ_t , w_e , w_g , prediction accuracy, and mean surprise from correction logs, comparing the first 5 corrections (episode start) against the last 5 (episode end) for episodes with ≥ 10 corrections. If self-calibration works as designed, prediction accuracy should rise, surprise should fall, σ_t should decrease from its initial 1.0, and exploration weight should increase—a behavioral shift emerging from surprise dynamics alone.

Table 4. Ablation results. (a) Correction: same configuration, only CORRECT phase differs. (b) Components added cumulatively to SCOPE. (c) Calibration metrics at episode start vs. end.

(a) Correction mechanism			
Variant	SR (%) ↑	N	Rel. Δ
Predict only (no correction)	12.75	345	—
Full PEC loop (ours)	29.74	343	+133%

(b) Component ablation			
Configuration	SR (%) ↑	SPL ↑	Δ SR
SCOPE baseline	—	—	—
+ Goal extraction	—	—	+—
+ Semantic predictor	—	—	+—
+ Uncertainty selection	—	—	+—
+ Online correction (full)	29.74	20.88	+—

(c) Within-episode calibration		
Metric	Start (first 5)	End (last 5)
Prediction accuracy (%)	—	—
Mean surprise (KL)	—	—
σ_t (confidence)	1.0	—
w_e (exploration wt.)	0.30	—
w_g (goal wt.)	0.20	—

4.5. Analysis

Why SPL matters more than SR here. PEC-Nav’s overall SPL of 20.88 may appear modest, but it reflects the exploration cost during early episode steps when the agent’s model is uncalibrated. The SPL restricted to successful episodes is **70.22**—meaning that once calibration converges, the agent navigates efficiently. This gap is *predicted by the theory*: the PEC loop pays an upfront exploration cost that is amortized as surprise decreases and the agent shifts from broad exploration to targeted search.

Small model, smart search. PEC-Nav uses a 4B-parameter VLM on a single consumer GPU, yet matches or exceeds methods using GPT-4V (~1.8T parameters via API). The implication is practical: self-calibrating search is a compute-efficient alternative to model scaling. Rather than throwing a larger model at imperfect predictions, PEC-Nav detects and corrects prediction failures online.

The predictor alone hurts. This is the most counterintuitive finding. Adding a semantic predictor to SCOPE *without* the correction loop degrades SR from the SCOPE baseline to 12.75%. The predictor’s overconfident—and often wrong—layout predictions steer the agent toward frontiers that match the prediction rather than the reality. Only when the correction loop is active does the system learn which predictions to trust, transforming a liability into the system’s greatest strength.

5. Conclusion

We introduced PEC-Nav, a self-calibrating frontier search mechanism for embodied question answering. The core insight is that prediction errors are more valuable than predictions: by measuring the surprise between anticipated and observed layouts, the agent adapts its confidence and exploration-exploitation balance within a single episode. On LMEE-Bench, PEC-Nav achieves 29.74% SR and 20.88 SPL with a 4B VLM on one GPU. The correction ablation—where disabling the feedback loop drops SR to 12.75%—provides the cleanest evidence that *closing the loop* is what matters.

Limitations. The CORRECT phase relies on the Habitat simulator’s ground-truth semantic sensor; replacing it with a learned detector is needed for real-world deployment. The predictor uses prompt-based inference without parameter updates—test-time adaptation could accelerate calibration. Extension to multi-floor environments and persistent cross-episode memory remains open.

References

- [1] Peter Anderson, Angel Chang, Devendra Singh Chaplot, Alexey Dosovitskiy, Saurabh Gupta, Vladlen Koltun, Jana Kosecka, Jitendra Malik, Roozbeh Mottaghi, Manolis Savva, and Amir R. Zamir. On evaluation of embodied navigation agents. *arXiv preprint arXiv:1807.06757*, 2018.
- [2] Anonymous. CoV: Chain-of-view prompting for embodied question answering. *arXiv preprint*, 2026.
- [3] Jinze Bai et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.
- [4] Devendra Singh Chaplot, Dhiraj Gandhi, Abhinav Gupta, and Ruslan Salakhutdinov. Object goal navigation using goal-oriented semantic exploration. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 4247–4258, 2020.
- [5] Yifan Cheng et al. UniWM: A unified memory-augmented world model for embodied navigation. *arXiv preprint*, 2026.
- [6] Abhishek Das, Samyak Datta, Georgia Gkioxari, Stefan Lee, Devi Parikh, and Dhruv Batra. Embodied question answering. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1–10, 2018.
- [7] Samir Yitzhak Gadre, Mitchell Wortsman, Gabriel Ilharco, Ludwig Schmidt, and Shuran Song. CLIP on Wheels: Zero-shot object navigation as object localization and exploration. In *Conference on Robot Learning (CoRL)*, 2023.
- [8] Yuncong Huang et al. 3D-Mem: 3d scene memory for embodied exploration and reasoning. *arXiv preprint arXiv:2411.17735*, 2024.
- [9] Yinuo Jiang et al. Fine-EQA: Fine-grained embodied question answering. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025.
- [10] Mukul Khanna, Naoki Yokoyama, Dhruv Batra, et al. GOAT-Bench: A benchmark for multi-modal lifelong navigation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [11] Yuxuan Nie et al. WMNav: Integrating vision language models into world models for object goal navigation. *arXiv preprint arXiv:2503.12907*, 2025.
- [12] OpenAI. GPT-4o system card. *arXiv preprint arXiv:2410.21276*, 2024.
- [13] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, et al. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning (ICML)*, pages 8748–8763, 2021.
- [14] Allen Z. Ren et al. Explore-EQA: Active embodied question answering with vision-language models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [15] Saumya Saxena et al. GraphEQA: Using 3d semantic scene graphs for real-time embodied question answering. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.
- [16] Bowen Sun, Jiahao Chen, Stefan Leutenegger, Cesar Cadena, Marc Pollefeys, and Hermann Blum. FrontierNet: Learning visual cues to explore. *arXiv preprint arXiv:2501.04597*, 2025.
- [17] Jiaqi Wang et al. Expand your SCOPE: Semantic cognition over potential-based exploration for embodied visual navigation. In *AAAI Conference on Artificial Intelligence*, 2026.
- [18] Jiaqi Wang et al. MemoryExplorer: Multi-goal exploration with episodic memory for embodied agents. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
- [19] Yuxuan Wang et al. ToolEQA: Tool-augmented embodied question answering. *arXiv preprint arXiv:2504.14202*, 2025.
- [20] Brian Yamauchi. A frontier-based approach for autonomous exploration. In *IEEE International Symposium on Computational Intelligence in Robotics and Automation (CIRA)*, pages 146–151, 1997.

756	[21] Shunyu Yang et al. RA-Mem: Retrieval-augmented memory for embodied agents. <i>arXiv preprint</i> , 2024.	810
757		811
758	[22] Hang Yin, Xiuwei Wang, et al. UniGoal: A universal zero-shot goal-oriented navigation framework. In <i>IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)</i> , 2025.	812
759		813
760		814
761		815
762	[23] Hang Yin, Xiuwei Wang, Miao Shi, et al. SG-Nav: Online 3d scene graph prompting for llm-based zero-shot object navigation. In <i>Advances in Neural Information Processing Systems (NeurIPS)</i> , 2024.	816
763		817
764		818
765		819
766	[24] Naoki Yokoyama, Dhruv Batra, and Sehoon Ha. VLFM: Vision-language frontier maps for zero-shot semantic navigation. In <i>IEEE International Conference on Robotics and Automation (ICRA)</i> , 2024.	820
767		821
768		822
769	[25] Yuxin Zhai et al. Memory-centric embodied question answering. <i>arXiv preprint arXiv:2505.10542</i> , 2025.	823
770		824
771	[26] Hanqing Zhang et al. Imagine before go: Self-supervised generative map for vision-and-language navigation. In <i>IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)</i> , 2024.	825
772		826
773		827
774		828
775	[27] Xinyu Zhao et al. ImagineNav: Prompting vision-language models as imagination for goal navigation. <i>arXiv preprint arXiv:2408.09992</i> , 2024.	829
776		830
777		831
778	[28] Kaiwen Zhou, Kaizhi Zheng, Connor Pratt, Noah Snively, and Ruslan Salakhutdinov. ESC: Exploration with soft commonsense constraints for zero-shot object navigation. In <i>International Conference on Machine Learning (ICML)</i> , 2023.	832
779		833
780		834
781		835
782		836
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